

Coding & Solution

A Comprehensive Measure of Well-Being | HDI Predictor

HDI Computation (TypeScript)

```
function computeHDI(life, mys, eys, gni) { const lifeldx = (life-20)/(85-20); const eduldx = (mys/15 + eys/18)/2; const incldx = (Math.log(Math.max(gni,100))-Math.log(100))/(Math.log(75000)-Math.log(100)); const clamp = n => Math.min(1,Math.max(0,n)); return Math.cbrt(clamp(lifeldx)*clamp(eduldx)*clamp(incldx)); }
```

Flask Predict Route (Python)

```
@app.route('/predict', methods=['POST']) def predict(): input_features = [float(x) for x in request.form.values()]; features_value = np.array(input_features); output = model.predict([features_value]); y_pred = round(output[0], 2); return render_template('resultnew.html', prediction_text=classify(y_pred))
```

Key Files

- src/routes/index.tsx — Main HDI Predictor component
- Flask/app.py — Flask backend with ML model
- Flask/templates/ — Jinja2 HTML templates
- docs/ — All phase documentation

Git Commits

- 55 structured commits across all epics
- Follows epic-wise commit convention
- Each commit maps to a specific task